DATA SCIENCE REPORT

PRODUCT TAXONOMY INTELLIGENCE: NLP-BASED PRODUCT CLASSIFICATION & EXPLAINABILITY

Character-Level Feature Engineering, Machine Learning & Classification Performance



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

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Overview & Business Objective

1.1 PROJECT OVERVIEW

E-commerce platforms and product comparison services manage large catalogs containing thousands of product listings from different merchants. These listings often vary in naming conventions, technical specifications, brand references, and product descriptions.

Manually categorizing such products can be time-consuming and inconsistent, particularly when similar products are listed under different names.

The Product Taxonomy Intelligence project applies Natural Language Processing (NLP) and machine learning to automatically classify products into predefined categories using their titles.

The project covers data exploration, text preprocessing, TF-IDF feature engineering, classical machine learning, TextCNN experimentation, model evaluation, error analysis, and explainability.



1.2 PROBLEM STATEMENT

Product titles frequently contain abbreviations, technical specifications, model identifiers, and overlapping terminology, making accurate classification challenging.

The key problem addressed in this project is:

How can machine learning classify products into predefined categories using product titles while maintaining strong performance across different product types?

The project formulates this as a 10-class supervised text classification problem, with *Category Label* as the target variable.

1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Analyze product listings and identify category-specific text patterns.
- Prepare product titles for NLP-based classification.
- Develop and compare word-level and character-level TF-IDF representations.
- Train and optimize multiple machine learning classifiers.
- Evaluate a TextCNN architecture as a deep learning alternative.
- Prevent information leakage through group-aware data splitting and cross-validation.
- Assess classification performance and investigate prediction errors.
- Interpret model decisions using character-level feature contributions.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, NLP-based feature engineering, machine learning classification, model explainability, and business interpretation to identify product naming patterns, evaluate classification performance, understand model decisions, and explore opportunities for automated product categorization.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Category distribution analysis
- Merchant and cluster analysis

02

NLP & Classification

- Text preprocessing
- TF-IDF feature extraction
- Group-aware validation
- Machine learning & TextCNN modeling

03

Model Explainability

- Character feature importance
- Misclassification analysis
- Decision margin analysis

04

Business Interpretation

- Automated product categorization
- Product catalog consistency
- Ambiguous listing identification
- Classification limitations
- Recommendations for improvement

Data Understanding & Exploratory Data Analysis

2.1 DATA OVERVIEW

The dataset provides a comprehensive view of product listings across multiple merchants and categories, covering product titles, identifiers, category labels, and product clustering information. These variables provide the foundation for analyzing product naming patterns, identifying category-specific terminology, and developing NLP-based machine learning models for accurate product classification.

Data Source: Kaggle

Dataset Size

35,311
Records

16
Original Features

Data Diversity

10
Categories

306
Merchamts

12,849
Clusters

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Product ID	Categorical (Identifier)	Unique listing identifier
Product Title	Text (Unstructured)	Textual product name
Merchant ID	Categorical (Identifier)	Identifier of the merchant
Cluster ID	Categorical (Identifier)	Product grouping identifier
Cluster Label	Text (Categorical Label)	Product grouping description
Category ID	Categorical (Nominal)	Numerical product category identifier
Category Label	Categorical (Nominal)	Product category name

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Demand

Category Label represents the product category that the model attempts to predict based on the textual information contained in each product title.

The target has:

10
Unique Categories

2,212
Smallest Category Count

5,501
Largest Category Count

15.58%
Largest Category Share

6.26%
Smallest Category Share

The distribution is moderately imbalanced, with Fridge Freezers representing the largest category and Freezers the smallest.

2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Temporal information
- Potential outliers
- Feature relationships

The assessment identified:

- 35,311 observations and seven variables.
- No missing values across the original columns.
- No fully duplicated rows.
- 30,993 unique product titles, indicating repeated titles across listings.

Following data preparation, 35,310 observations were retained for modeling.

Repeated product titles and product clusters were important considerations when designing the train-test split to reduce information leakage.

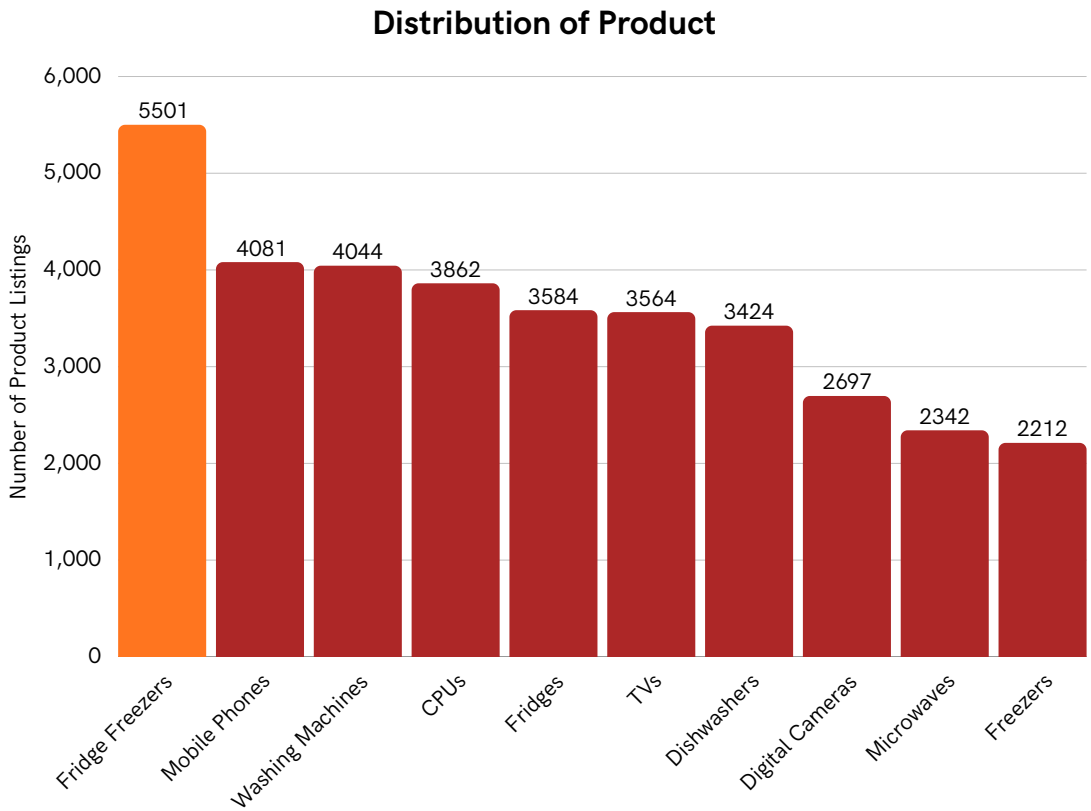


2.4 Exploratory Data Analysis

Product categories are moderately imbalanced, with Fridge Freezers representing the largest share of listings.

15.58%

Fridge Freezer Listings



Key observations

- Fridge Freezers account for approximately 15.58% of listings, representing the largest category.
- Freezers account for approximately 6.26%, making them the least represented category. The remaining categories have relatively moderate differences in representation.

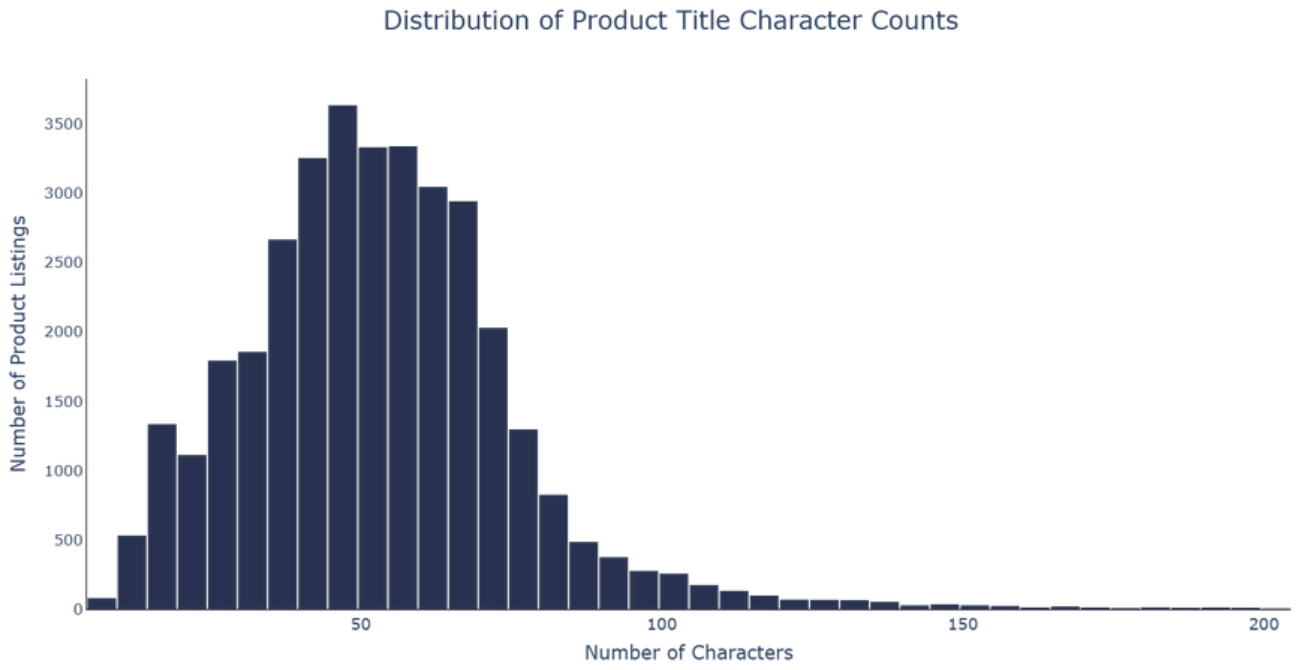
Business Insights

- Category-level evaluation is necessary to identify performance differences that overall accuracy may conceal.
- Macro-averaged metrics provide a useful assessment of classification performance across categories.

Product titles vary considerably in length, averaging approximately 53 characters and nine words per listing.

53

Average Title Length



Key observations

- Product titles contain approximately 53 characters and 8.67 words on average.
- Title lengths range from 5 to 200 characters.
- Product descriptions vary considerably in the amount of information provided.

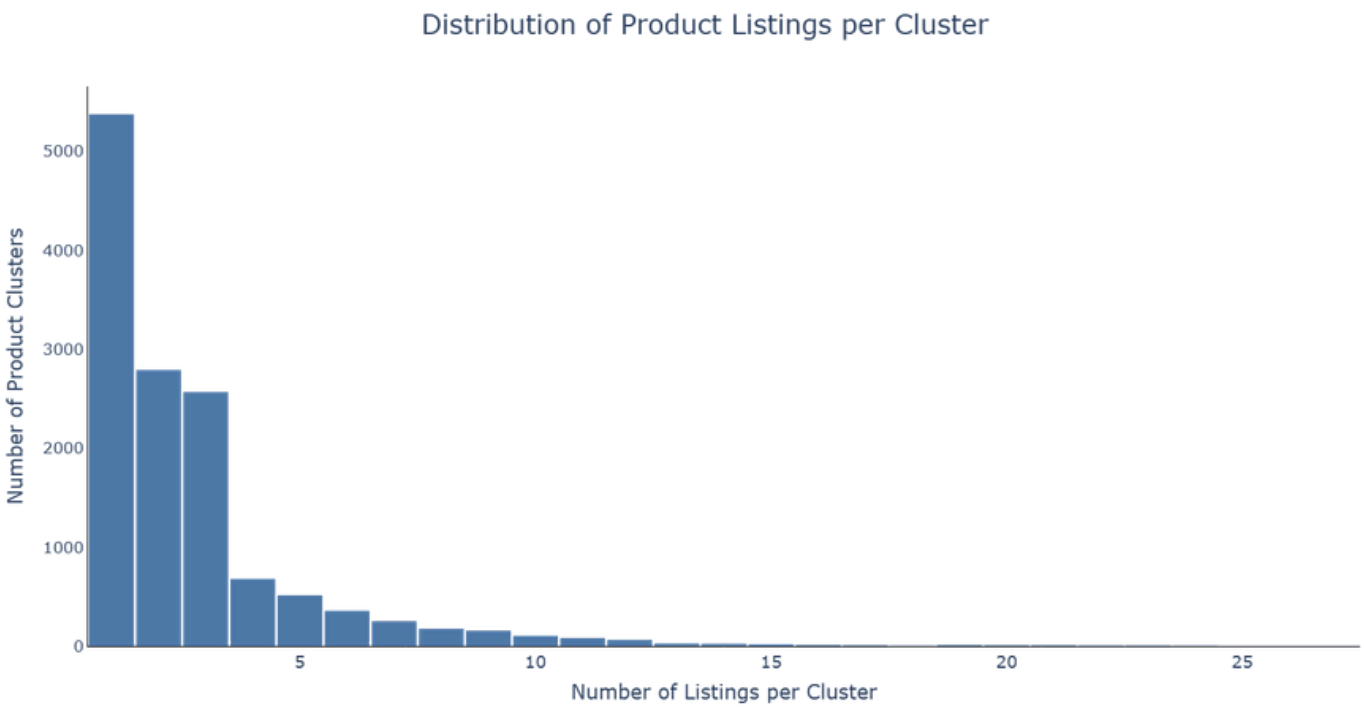
Business Insights

- Short titles may provide limited evidence for classification.
- Character-level representations can capture useful patterns within abbreviated names and model identifiers.

Most product clusters contain multiple listings, indicating repeated representations of underlying products.

59.39%

Multi-Listing Cluster Share



Key observations

- The dataset contains 13,233 unique product clusters. Of these, 7,859 clusters contain multiple listings.Individual clusters contain up to 27 listings.

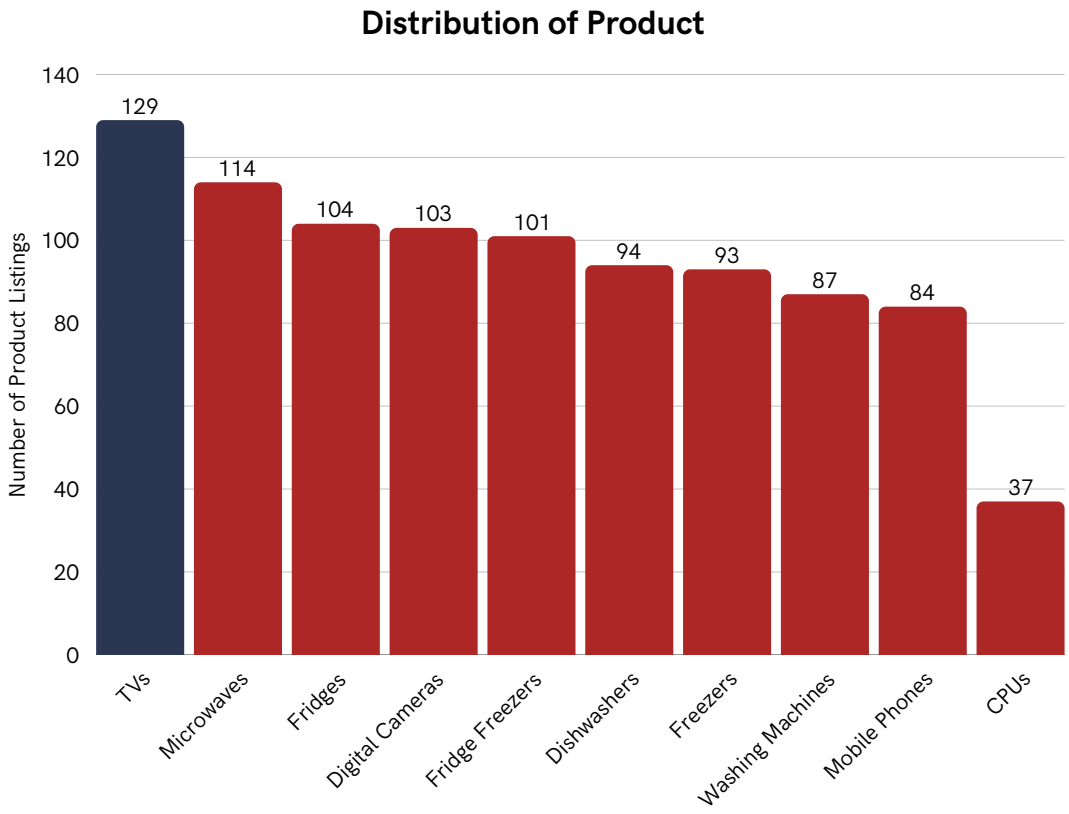
Business Insights

- Multiple listings may refer to the same underlying product.
- Product grouping information must be considered during validation to avoid overly optimistic performance estimates.

Product listings are unevenly distributed across merchants, with the largest merchant contributing substantially more listings.

2,547

Largest Merchant Listings



Key observations

- The dataset contains 306 unique merchants. Merchant representation varies substantially, with the largest merchant contributing 2,547 listings.
- Some merchants specialize in a single category, while others list products across multiple categories.

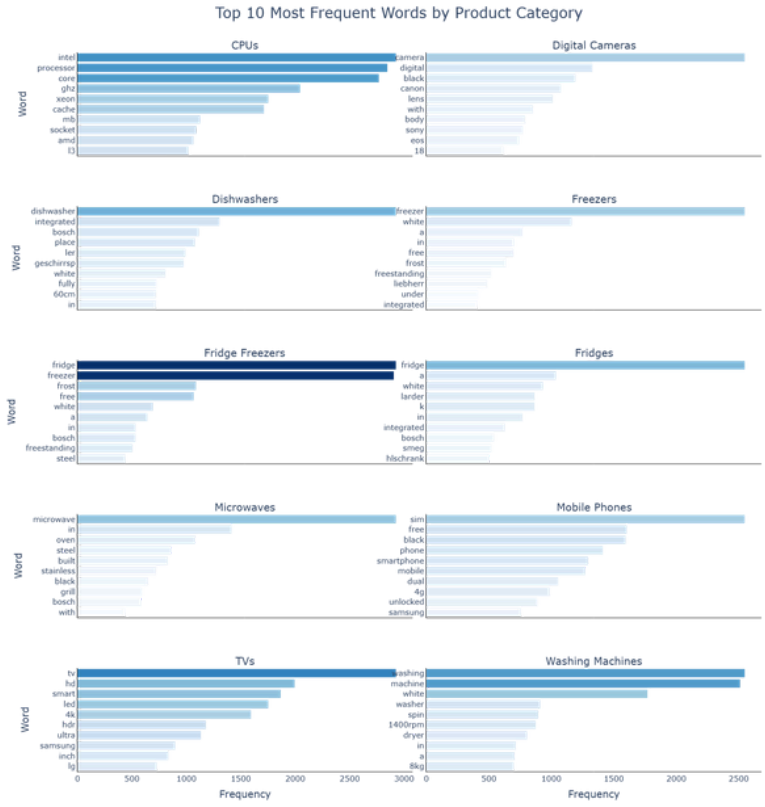
Business Insights

- Product naming patterns may vary between merchants.
- A robust classification system should learn transferable product terminology rather than depend exclusively on merchant-specific patterns.

Product categories exhibit distinctive terminology, although related appliance categories share overlapping keywords.

10

Product Categories Analyzed



Key observations

- Product categories exhibit recognizable terminology associated with their functions and technical specifications.
- Common terms include fridge, freezer, intel, processor, washing, and smart.
- Related categories share terminology, particularly Fridges, Freezers, and Fridge Freezers.

Business Insights

- Text-based classification is feasible because product titles contain meaningful category information.
- Overlapping terminology may introduce ambiguity between closely related categories.

Data Preparation & Feature Engineering



3.1 DATA PREPARATION

Following exploratory analysis, the dataset was prepared for supervised text classification.

The preparation process included:

- Cleaning and standardizing the dataset.
- Selecting Product Title as the primary predictive input.
- Separating the target variable.
- Identifying related product listings.
- Creating group-aware training and testing partitions.
- Encoding target categories.
- Transforming product titles into numerical representations.

The resulting dataset contained 35,310 observations.



3.2 FEATURE SELECTION & LEAKAGE PREVENTION

The model uses product titles as its primary input rather than relying on product identifiers, merchant identifiers, or category-related metadata.

Identifiers and grouping information were considered during data preparation to reduce leakage between related product listings.

This approach encourages the classifier to learn patterns from product names rather than memorize dataset-specific identifiers.



3.3 TRAIN-TEST SPLIT

The dataset was divided using a group-aware strategy.

Dataset	Observations
Training	28,919
Testing	6,391
Total	***35,310***

The test dataset was reserved for independent evaluation.

Group-aware splitting was used to reduce the risk of closely related product listings appearing across training and testing partitions.

3.4 TEXT FEATURE ENGINEERING

Product titles were transformed into numerical representations using ***Term Frequency–Inverse Document Frequency (TF-IDF)***.

The experimentation considered word-level, character-level, and combined feature representations.

Word-Level TF-IDF

Captures individual words and word combinations that describe product categories, brands, and specifications.

Character-Level TF-IDF

Captures shorter character sequences within product titles.

This representation is particularly useful for:

- Model identifiers and technical specifications.
- Abbreviated product names.
- Variations in product naming conventions.
- Partial words and recurring character patterns.

Combined TF-IDF

Combines word-level and character-level information to evaluate whether complementary representations improve classification.



3.5 TARGET ENCODING

The 10 product categories were encoded into numerical labels for model training and evaluation.

The fitted label encoder was retained to ensure consistent category mappings across subsequent notebooks.

3.6 FINAL FEATURE REPRESENTATION

The selected model uses 50,000 character-level TF-IDF features.

Component	Final configuration
Input	Product Title
Target	Category Label
Classification classes	10
Feature representation	Character TF-IDF
Feature dimensions	50,000
Training observations	28,919
Testing observations	6,391

